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Emergency Department Performance and Poverty Levels Across States

Abstract

This study explores the relationship between poverty rates and emergency department (ED) performance scores in the United States at the state level using public datasets from the Centers of Medicare & Medicaid Services (CMS) and the U.S. Census Bureau's American Community Survey (ACS). States were classified into "Higher" and "Lower" poverty groups based on the median poverty rate of the states in the sample. ED performance scores were aggregated of data to the state level and merged with state level poverty estimates. The applicability of the analysis is limited by data availability and aggregation to the state level.

1. Introduction

Access to timely and effective emergency care is a key component in a healthcare system and is vital in communities. Prior studies and research indicate that socioeconomic factors, such as income and insurance coverage, affect access to proper healthcare, preventing communities from receiving essential treatment.

Performance scores from hospital EDs, which represent standardized median wait time-based metrics, may reflect differences in access across communities with varying economic conditions.

This study examines whether poverty levels within states affect emergency department performance. By aggregating hospital performance data to the state level and using poverty estimates from the ACS, this analysis provides a look at how socioeconomic backgrounds correlate with health system performance scores. This study aims to contribute to a better understanding of health equity and to examine disparities between states with different economic statuses.

2. Data Sources

Emergency Department Performance Data: Data on hospital emergency department performances were taken from the Centers for Medicare & Medicaid Services (CMS) "Timely and Effective Care" dataset. This dataset includes standardized performance scores for ED measures for individual hospital facilities across the United States.

Performance scores for the emergency department represent median wait times in EDs in minutes.

3. Methods

Data Cleaning

Hospital performance data was cleaned first by dropping all records with unavailable scores. The cleaned hospital dataset was then aggregated to the state level by computing the mean performance score (`avg_er_score`) across all hospitals in each state. The number of hospitals per state was also recorded. Poverty data was cleaned in a similar way by dropping variables besides percentage of families below the poverty line and geographic identifiers.

Merging and Grouping

State-level poverty estimates were determined by obtaining the mean percentage of families below the poverty line in all counties available within the state. States were then classified into “Higher Poverty” if the mean poverty percentage was above the median of all states and “Lower Poverty” otherwise.

Analysis

Comparisons were drawn by computing group averages of ED scores and visualizing score distributions using boxplots. Boxplots were chosen to illustrate variation and overlap between groups. Additionally, a scatterplot was made to see correlation between poverty rates and emergency department scores, and a correlation coefficient was added to accurately determine the correlation. All analyses were performed using Python with the pandas library for data manipulation and matplotlib for visualization.

4. Results

The merged dataset included state-level measures for both average emergency department performance and average poverty rates. Boxplot comparisons showed states with a higher than median poverty exhibited slightly higher average ED performance scores than those with less poverty. Some overlap of ED scores were also shown between the two groups (Figure 1).

To further examine the relationship of poverty and ED performance, a scatterplot was constructed with the same data. The scatterplot revealed poverty rates and ED performances did not have a strong linear relationship at the state level. Pearson correlation analysis showed a weak relationship between poverty rate and average ED performance score ($r \approx -0.31$). This suggests that poverty rate alone is not a predictor of state-level ED performance.

5. Discussion

Interpretation of Findings

In this sample, states with higher poverty rates exhibited higher average ED performance scores when grouped by median poverty level (Figure 1). This counterintuitive result could be explained by a multitude of factors. Hospitals serving lower-income populations may receive additional funding through safety-net funding programs, which could correlate with better performance scores.

Although on average states with higher poverty demonstrated better average ED scores, scatterplot analysis revealed no strong linear relationship between poverty rate and ED performance scores (Figure 2), suggesting that poverty does not necessarily correlate with ED performance scores. However, these findings should be interpreted cautiously given the limited number of states in the sample and the aggregation of data at the state level, which may mask variation within states.

Limitations

This study has multiple limitations. Firstly, the number of states in the study was limited (14 states) due to the lack of performance scores listed in some states, which reduces generalizability. Secondly, the analysis was conducted at the state level, which could hide important variation based on socioeconomic status. For example, even if a state was grouped as "Lower Poverty", there could be separate regions with varying poverty levels with different ED performance score patterns. Finally, only one variable was examined. In other words, there could be very little correlation between poverty levels and ED performance scores and other variables such as hospital staffing or patient volume are the things that actually contribute, rather than socioeconomic status.

Implications

Despite these limitations, the approach illustrates how publicly available data can be used to investigate health equity questions. Future work could incorporate more data with better generalizability to lower the limitations of studies.

6. Conclusion

This study explored how poverty levels correlated with ED performance scores. When grouped by median poverty level, states with higher poverty appeared to perform better than those with less poverty. However, scatterplot analysis showed little correlation between poverty rate and ED performance scores. These findings suggest factors other than poverty are most likely what impact emergency department performance scores. Future research at a more granular level may be able to provide a clearer understanding of how socioeconomic conditions influence emergency care by removing some limitations.

7. Figures

Figure 1

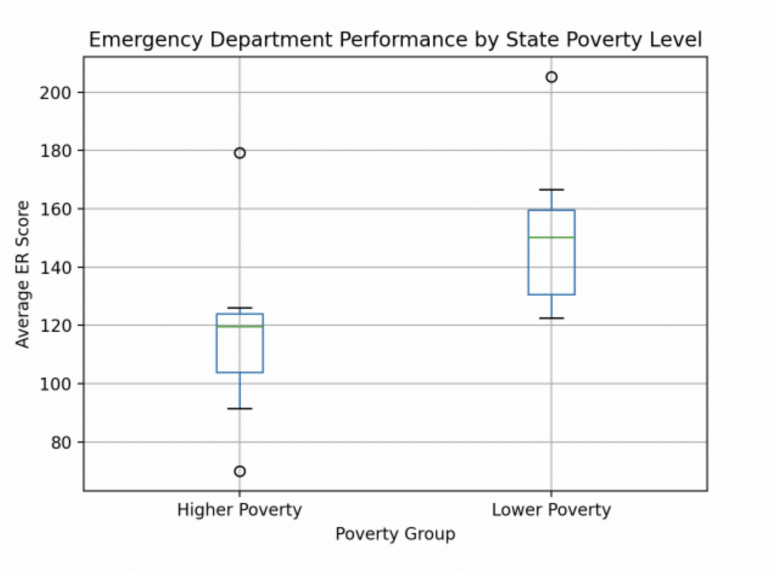
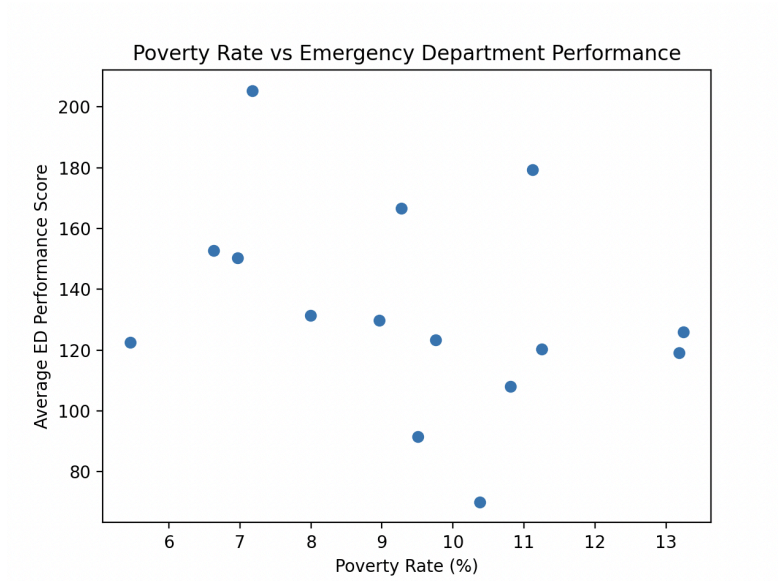


Figure 2



8. References

Centers for Medicare & Medicaid Services. *Timely and Effective Care – Emergency Department Data* (CMS Hospital Compare).

U.S. Census Bureau. *American Community Survey (ACS) 1-Year Estimates, Table S1702: Poverty Status in the Past 12 Months*.